Dismantling and remounting

Dismantling of the Cover Guard:

Remove the side hatches first and then the top section.

Clean the covers.

Rinse the press cage before stripping. (This should always be done after the press has been put out of commission to prevent clogging of the holes in the cage plates).

Dismantling and Remounting of Upper Part of the Press Cage:

The press cage consists of an upper part and a lower part arranged as mentioned in the above description and shown on drwg. No. 41-35247. Remove the upper stay bar at the discharge outlet.

The upper part of the cage is split vertically in the middle. To remove the two parts of the cage, swing down the forked sling clamps and loosen the screws from the split bridge. The "Outlet End Part" can now be lifted off.

If it is desired to lift off the inlet end part of the cage, the other part must be loosened and moved away from the inlet chute before lifting off. Alternatively the inlet chute may be taken off, which also gives the necessary space.

When remounting the press cage, check that the bridges in the upper and lower part correspond accurately. The stay bar should be tightened so that it is in compression and no clearance is to be found. See that the surfaces of the side bars are clean, tighten well all the clamping screws, and ensure that there is no clearance between the upper and lower side bars.

Removal of Inlet Chute:

When the press inlet chute is to be removed, some free space above the press chute is needed for lifting. (Minimum 470 mm - 19 inches). Disconnect top flange of the chute, and remove the lower flange screws away from the side bars and the end plate. Lift the chute carefully away. When remounting, clean adjoining surfaces and see that the chute fits well and is clear of press screws before tightening the fixing screws.

Removal and Remounting of Backing Plates:

In order to take out the plates from the cage it is necessary first to unscrew the side bars from the bridges to the plates in question (see drwg. 41–35247). The plates are then loose and can be removed. It is not necessary to loosen the center bar.

We advise to mark each plate when removing, in order to ensure correct refit.

The plates in the lower part of the cage may be removed without removing the press screws. A suitable crowbar may be useful to get hold of the backing plates and raise them, as indicated by dotted lines in drwg. 41–35247. Be careful that the point of the crowbar does not damage the strainer plate. Take care also of the arched distance pieces between the backing plates.

When a frame under the feeding chute is to be taken out, the chute has to be removed and the upper cage, if not already dismantled, moved along sufficiently to free the side bars.

The outer surfaces of the backing plates are machined accurately to the diameter of the corresponding bridges, and it is consequently of great importance that they are placed in the original position when reassembling. Check by means of a gauge that there is no clearance between backing plates and bridges.

When fixing the side bars, make sure that the screws are pulled tight. Some gentle blows with a soft mallet near the screws at the same time as the screws are tightened, may be effective.

Removal of Strainer Plates:

The strainer plates are spot–welded or riveted to backing plates. They are usually spot–welded to backing plates of stainless steel, drawing No. 41–34405, and riveted to mild steel backing plates, drawing No. 41–33432.

When replacing a strainer plate spot welds are broken with a flat chisel, rivets are drilled out.

When the strainer plate is removed, the backing plate ought to be cleaned with a wire brush and washed. Remove remaining spots by careful grinding.

Replacement of Strainer Plates:

Spare strainer plates should always be kept in stock at the factory. On request, however, such plates are supplied at short notice, cut to accurate size and perforated as prescribed, but not rolled.

The strainer plates are spot–welded or riveted to backing plates.

Spot–welding (for stainless backing plates only):

The strainer plate should be preformed by rolling and clamped to the backing plate in a simple fixture, as shown on drwg. 41–34405, after which spot–welding takes place.

The clamping fixture consists of two semi–circular pieces of wood with the same radius as the press screw + 1 millimetre, 2 tension girths with threads and nuts, and 2 cross bars. The figure shows the assembly of the plates in the fixture.

Before welding check that the strainer plate is in the correct position on the backing plate.

The first spot weld should be located nearest to the middle of the arched edges of the plates – in both ends, proceed towards the edges.

When the welding has been done, bend the protruding edges of the strainer plate over the backing plate. At the arched ends the strainer plate should be split between the welds to facilitate bending – see drawing. At each of the four corners a small superfluous piece of the plate has to be cut away. The edges should be hammered thoroughly.

Finally the surplus material of the strainer plate edges must be well trimmed.

Replacement of strainer plates on the frame positioned under the feeding chute, is carried out in the same way as the above procedure.

The fastening of strainer plates can alternatively be carried out by tackwelding, preferably by TIG–welding.

Riveting (for stainless or mild steel backing plates):

Strainer plates can also, as by the previous procedure, be fastened by means of rivets after drilling and counter sinking of rivet holes in backing plates, see drawing 41–33432.

The rivet holes in the strainer plate should preferably be made by means of a drill, having a diameter a little smaller than that of the rivet. After drilling, countersinking takes place as indicated on drawing 41–33432. The holes might also be made by punching with a pointed drift. By this method, however, there is a risk that the strainer plate may rise around the hole during the riveting procedure.

Hammer the strainer well in order to avoid clearance between backing and strainer plate.

Surplus material of the rivet heads and the strainer plate edges must be well trimmed.

Removal of Press Screws:

Ref. main drawing.

When handling the press screws and other heavy parts of the press, it is important that the lifting tackle is sufficiently powerful. The weight of the different parts is noted on the special list provided.

When the inlet chute and upper parts of the press cage have been removed as described previously, remove the fixing screws from the flanges of the gearbox output shafts. Flanges and screw bodies have corresponding marks to facilitate later reassembling. If these marks are illegible, marks should be renewed before dismantling is carried out.

Inlet end plate and cover plate for frame are removed and the connection between bearing houses and carrier undone. Take care of seal house, end plate/supporting plate and bearing housing/carrier.

Put a sheet of gasket or similar material (approx. 3 mm thick) under the press screw, between the flights and the second strainer plate (counted from the inlet end). Slacken the hoist until the screws are resting on the gasket.

Loosen the gearbox output shafts from the press screws by means of the dismantling screws (supplied) by inserting these in the threaded holes of the shaft flanges. During this operation both forcing screws should be turned at the same rate. Take care of the dowels, which are fitted between flange and screw body.

When flange and press screws are disconnected, make sure that there is a clearance between the flange and the press screw before lifting the latter.

Place a safe cable rope sling around one of the press screws at its centre of gravity and lift it gently out of the press. Lowering should be done on a prepared place, preferably on wooden beds, avoiding damage to the screw flights.

All handling of the heavy parts should be arranged safely and with great care.

If bearings and seal house are to be dismantled from the shaft, this can be done after the press screws have been removed from the press. Particular consideration to seals and gaskets must be exercised. To ease refitting, the positions on the journal should be marked before they are detached. The position of the roller bearings should also be marked. See that the seal rings are facing the right way when refitting.

Remounting of Press Screws:

It is presumed that bearings, housing and seal ducts are mounted on the press screw journal before refitting the press screw.

The press screw is lifted at its centre of gravity and lowered into the press cage so far that only a small clearance exists between bearing housing and frame.

The screw is then axially abutted to the gearbox output shaft flanges. The flights of one screw should run midway between the flights of the other.

The flange screw bolts are pulled to 2200 Nm (1660 ft.lbf) and locked by locking washers.

The end plate and seal house are then mounted. Remember the dowels for the end plate. The bearing housings are filled with grease according to lube chart.

When checking the fitting, turn the screws by turning the gearbox input shaft by hand some turns, then afterwards by motor.

Replacement of oil seals on gearbox output shafts:

All item Nos. refer to section drawing of gearbox and drawing No. 41–33429

The "Split Klozure" oil seal (item 44) can be replaced by loosening the screws on the oil seal cover (item 14) and the screws in the supporting ring (item 16) and pulling the cover and ring including the two V–rings away from the oil seal housing.

If any of the V–rings (item 45) are to be replaced, the press screws must be disconnected from the gearbox and pushed away from the flange about 80 mm (3"). Remember first to disconnect the end plate, seal ducts, retainer and the inlet end bearing as these will have to follow the screws rearwards.

The end plate must be lifted off and the seal ducts swung outwards. The press screws are displaced by means of dismantling screws in the flanges. Make sure the inlet bearing housings are free.

The upper cage outlet end should be taken off. The cover (item 14) is detached and the defective V–rings are cut off and removed. The new V–rings can be fitted to the output shaft by stretching until they slip over the flange and the covers (items 15 and 14). Make sure the sealing lip faces the right way. The V–rings should be oiled or greased and treated carefully in order to avoid damage.

The stretching and slipping over the flange should be carried out by two men, one on each side of the flange. A piece of rag should be wrapped around the V–ring, protecting this when sliding it over the edge of the flange. The V–ring should be stretched evenly.

A pair of wooden levers can be used to lever the V–ring over the flange.

After having replaced the V–ring the confines should be filled with grease. This is done by undoing both of the plugs in the cover (item 14) and fitting a standard grease nipple in one of the holes. Then grease is pressed in by a grease–gun until it flows out of the other hole. Now, refit the plugs. It is not necessary to lubricate during normal running, but it is recommended to change grease once a year.

Dismantling of Gearbox:

All item Nos. refer to section drawing of gearbox.

When the gearbox is to be dismantled, the screw connection between press screws and gearbox output shafts has to be disconnected. The screws are displaced rearwards about 35 mm freeing the guide on the flange from the screws. To displace the screws it is necessary at the inlet end to disconnect the seal ducts, end plate, bearings and seal retainers.

The end plate for the cover is to be removed.

The pipe from the oil pump, with gauge and filter is detached. The output shaft oil seal housing (item 13) is loosened from the gearbox, and is moved somewhat outwards on the shaft.

Seal retainers and seal covers (item 13 and 14) can be left assembled for the time being.

In recesses behind the seal boxes, 6 springs (item 31) are positioned. The function of these is to locate the outer bearings when the press is running idle. The springs are now loose. Bearing covers for the screw shafts should be taken off.

Screw bolts in the flanged connection between top and bottom part of the gearbox should be undone. (2 of the bolts are fitted bolts).

On the side facing the press the bolts are internally accessible through scuttles on top of the gearbox. The top section should be lifted gently away, avoiding damage to internal pipework etc. Then the two output shafts including flanges, spur wheels, gland houses and bearings can be lifted out. The bearings should be covered up. Bearing covers for shafts No. 3 are loosened. The screws in the flange between the gearbox mid-section and bottom section are undone (2 of the bolts are fitted bolts). Further, the bolts connecting gearbox and frame are removed (6 bolts). The mid section can now be lifted away.

The two shafts No. 3, complete with bearings and gear wheels, are removed. The bearings should be covered up. The shafts must be marked, ensuring repositioning in the previous location.

The two shafts No. 2 (item 24), can be lifted out complete with gear wheels – this when bearing covers, bearings and bearing housings have been removed. The shaft should be marked. For removal of input shaft, the bearing cover, bearing and bearing house have to be removed before the shaft can be taken out. To facilitate the dismantling, all shafts have grooves for oil pressure connection.

Assembly of Gearbox:

See drawings Nos. 41-31751 and 41-33429.

All machined surfaces on gearbox must be cleaned and traces of old gasket material should be removed. Roller bearings should be well cleaned. Mating flanges between the gearbox sections should be made leakproof by means of suitable gasket compound (nonhardening type Permatex No. 2 or similar), applied in a thin even layer. See that no gasket compound material gets into the bearings and the gearbox bearing recesses.

When fitting the gearbox main sections the fitted bolts should be watched to ensure correct fitting.

When the output shafts are put in place, the teeth of gear wheels should engage in such a way that the mutual position of press screws is correct. This is when the match-marks on the flange and screw coincide and the flights of one screw are running midway between the flights of the other.

Avoid damage to the V-rings (item 45) and the Split Klozure rings (item 44) during assembly. Then seal housings, covers, support rings and seals should be taken apart and fitted individually. See drawing No. 41-33429. Remember the springs (item 31).

The confines around the V-rings should be filled with grease. See section about replacement of seals on the gearbox output shafts.

Fitting of Roller Bearings:

Most of the bearings in the gearbox and the inlet end bearings are mounted on tapered sleeve.

When fitting roller bearings by means of tapered sleeve, it is of utmost importance that the fitting of the sleeve is done correct.

The reduction of radial clearance (clearance between rollers and the race) is an expression for how tight the bearing is fitted.

It is therefore necessary to measure the clearance before and after the fitting of the sleeve, and check that the reduction of clearance is according to tabulation given in the end of this section.

Before measuring, the bearing should be turned a few times to make sure the rollers are in position. Further the bearing should be free on the shaft without external loads when the mea- surement is conducted.

The radial clearance is measured by means of feeler gauge pulled hard through the bearing.

Alternatively axial displacement of sleeve can be found. However, this method is unprecise and is only recommended when accessibility prevents the radial clearance to be measured.

The tapered sleeve is driven in by considerate blows by hammer on the sleeve rim (item 19 and 65). Use a piece of hard wood between hammer and sleeve.

SKF or Stord International A/S can supply hydraulic equipment for dismantling and refitting of tapered bearing sleeves.

The following tabulation shows the required clearances and the corresponding axial displacement when fitting SKF spherical roller bearings with tapered inner race.

Metric:

Bore diameter mm	Reduction of radial clearance mm	Axial dis- placement taper 1/12 min mm	max mm	Min. remaining clearance mm	Bearing designation
80–100	0.045–0.060	0.70	0.90	0.035	22220 CK
100–120	0.050–0.070	0.75	1.10	0.050	22224 CK
120–140	0.065–0.090	1.10	1.40	0.055	23128 CK

Imperial:

Bore diameter	Reduction of radial clearance	Axial dis- placement taper 1/12 min	max	Min. remaining clearance	Bearing designation
3⅛" – 4"	0,0018"–0,0025"	0,027"	0,035"	0,0014"	22220 CK
4" – 4¾"	0,0020"–0,0028"	0,030"	0,045"	0,0020"	22224 CK
4¾" – 5½"	0,0025"–0,0035"	0,045"	0,063"	0,0022"	23128 CK

Removal and refitting of shafts in the lower section of gearbox

All items Nos. refer to section drawing of gearbox.

General:

The three shafts in the lower section can be removed and replaced without having to dismantle the upper sections or shafts.

Before starting the dismantling, all bearing covers should be marked to ensure correct repositioning. Outlet chute for press cake will have to be removed to allow accessibility to the bearings on the output side of the gearbox.

Removal of input shaft (item 25):

Both of the bearing covers are removed. The washer on the rear side of the gearbox (item 19) is detached, and the tapered sleeve is pulled off by tightening up the removal nut (SKF KM 22) (delivered with the press). Bearing covers (item 9) inclusive bearing and snap ring are taken away.

The shaft inclusive of bearing (item 36) can now be pulled out. This can be done together with the bearing housing (item 8).

Removal of shaft No. 2 (item 24):

See drawing No. 41–31767.

Gearwheels (items 29 and 30) must be well supported against the bottom of the gearbox by means of wooden props.

Bearing covers, end washers, tapered sleeves, bearings and bearing houses in both ends should be detached as described for the input shaft. The stopper for the key (item 81) is removed.

The gearwheel is well propped against the gearbox wall by means of two wooden blocks and the shaft is pressed out through the gearwheel hub as illustrated by drawing No. 41–31767.

Insert and enter the extraction bolts through a thick steel plate which has a jack between this and the end of shaft (shown in principle on drawing No. 41–31767). The shaft may then be pressed out by means of the nuts or the jack. The two shafts have to be taken out on alternate sides of the gearbox. Seen gearbox face, the right shaft will have to come out through the rear side and the left through the front.